

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MRBENCH | Context: Metropolitan India — Grades 1–8 Mathematics Tutoring (Student Population)

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Justification

MRBench's eight-dimension pedagogical taxonomy is rigorously derived from learning-sciences principles [Q1, Q15, Q129] and covers core mistake-remediation behaviors (mistake identification, guidance, not revealing the answer, tone, human-likeness) that are relevant to any tutoring deployment. However, for the Indian metropolitan Grade 1–8 deployment — explicitly flagged HIGH priority and confirmed in elicitation A1 to require NCERT-aligned curriculum coverage — the taxonomy contains no category for curriculum alignment, Indian numeral conventions (lakhs/crores), rupee-denominated content, or culturally appropriate register. The authors themselves acknowledge the taxonomy 'will need to be verified on and likely adapted if applied to other tasks ... and to subjects other than mathematics' [Q118] but do not extend this to non-Western educational settings. Web search confirms no NCERT Grade 1–8 tutoring dialogue benchmark exists [WEB-6, WEB-7], so the gap is not addressable by alternative resources. The taxonomy provides partial but inadequate coverage of the categories required for this deployment.

ID	Evidence
WEB-6	No NCERT-aligned Grade 1–8 English-medium tutoring dialogue benchmark exists
WEB-7	HAWP is Hindi arithmetic word problems only, not tutoring dialogue

Strengths

- Pedagogically grounded eight-dimension taxonomy derived through iterative validation [Q15, Q129, Q132]
- Explicit focus on mistake remediation in mathematics matches the deployment's task scope [Q12, Q117]
- Covers two curriculum levels (elementary/Bridge and middle school/MathDial) corresponding to Grades 1–8 banding [Q143, Q144]

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required regional categories include NCERT-aligned topic coverage (place value with lakhs, fractions, integers, linear equations, mensuration), Indian numeral conventions, rupee-denominated word problems, and culturally appropriate register. Confirmed via elicitation A1 and NCERT syllabus sources [WEB-2, WEB-4, WEB-5].

ID	Evidence
WEB-2	NCERT Grade 1–5 mathematics syllabus (place value with lakhs, spiral sequencing)
WEB-4	NCERT upper-primary syllabus emphasizing abstraction, fractions, integers, algebra
WEB-5	NCERT Class 8 chapter list (Rational Numbers, Linear Equations, Mensuration)

IO-2: Yes. The benchmark's taxonomy omits curriculum alignment, Indian numeral conventions, and cultural-register categories. The authors acknowledge adaptation would be required for other educational settings [Q118, Q130], and no NCERT-aligned tutoring dialogue benchmark currently fills this gap [WEB-6, WEB-7].

ID	Evidence
WEB-6	No NCERT-aligned Grade 1–8 English-medium tutoring dialogue benchmark exists
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IO-3: The taxonomy's pedagogical dimensions (mistake identification, guidance, actionability, coherence, tone, human-likeness, etc.) are broadly relevant; no included category is clearly irrelevant to the Indian deployment, though the operationalization of Tutor Tone trivially collapses to non-offensiveness [Q90].

IO-4: Documented gaps: (a) no NCERT curriculum-alignment dimension; (b) no Indian numeral/currency convention dimension; (c) no cultural-appropriateness/register dimension. These omissions harm content validity for the target deployment because A4 confirms cultural appropriateness is a necessary condition for satisfaction.

Information Gaps

- No documentation of whether MRBench's source datasets contain any NCERT-aligned content
- No empirical study comparing pedagogical taxonomy coverage across Indian vs. Western tutoring contexts

Requires Expert Verification

- Whether additional pedagogical dimensions (e.g., respect/hierarchy register, scaffolding to NCERT learning outcomes) are required for the Indian metropolitan cohort

Remediation

Gap 1: Taxonomy omits NCERT curriculum alignment, Indian numeral convention compliance, and cultural-register categories required by the deployment.

Recommendation: Extend the eight-dimension taxonomy with at least three additional categories (curriculum-alignment, numeral/currency-convention compliance, cultural-register/warmth) and validate them with Indian educators before any deployment QA decisions are made on benchmark scores.